

A FORCE OS

Product Specification

Governing specification & architecture · 2026-08-01

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2. Constitution (v1.0, locked)

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AForce OS — Product & Technical Specification

Current as of June 23, 2026. Brand system v2.1.0. This document is compiled from the live codebase (`artifacts/aforce-os` , `artifacts/api-server` , `lib/`) and the locked product decisions in `replit.md` . Where the code and older spec PDFs disagree, the code is treated as the source of truth.

1. Product Overview

AForce OS is a real-time human-performance operating system delivered as a React Native / Expo mobile app backed by an Express 5 + PostgreSQL API server. Its core job is **hydration intelligence**: it turns logged fluid intake, biometrics, and environment into a single live performance score, then coaches the user with an AI voice engine. Around that core it layers recovery, social accountability, competition, and an integrated store/subscription business.

Positioning mantra: *Pause → Hydrate → Lock In → Perform*. **Tagline:** *Performance Is Non-Negotiable*.

2. Product Locks (non-negotiable design rules)

These constrain every feature and cannot be silently overridden.

- **Water-First Command System.** Recommendation order is **Water → Command → Optional support → Score update**. Default coach copy must begin with `HYDRATE NOW / Start with water` . Products never lead; behavior first, product second.
 - **Score Protection.** Only *completed* actions change the score. Recommendations, scans (HydroScan is advisory), and product selection never increase score. Every Home read-out is a read-only projection of already-completed behavior.
 - **Language / Localization Lock.** Launch languages: English, Spanish, French, German, Portuguese, Italian. No country-specific prioritization; architecture stays modular so hidden locales can ship later behind flags.
 - **Engine / UI Governance.** The engine may grow smarter; navigation may not grow. Feature flags control exposure. Principle: **Build 100% · Show 10% · Unlock over time**. One engine, multiple experiences.
 - **Product Positioning — Body First, Product Last.** *The body comes first. The recommendation comes second. The product comes last.* Decision order: Context → Recovery → Behavior → Learning → Optional support. Products support behavior; they never drive it — every recommendation is earned through intelligence, never sold. Applies ecosystem-wide, including Phantom Band™ and Meridian™. Canonical statement: `docs/PRODUCT_POSITIONING_PRINCIPLE.md` (governing: `AFORCE_OS_ARCHITECTURE_V1.md` → “Product Positioning Principle — Body First, Product Last”).
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3. Technical Architecture

Layer	Technology
Monorepo	pnpm workspaces
Mobile	React Native / Expo SDK 54, Expo Router 6, Reanimated 3, Gesture Handler
Mobile state	React Context + <code>useReducer</code> (slice-based store)
Data fetching	<code>@tanstack/react-query</code> with Orval-generated hooks
Backend	Node.js 24, Express 5, PostgreSQL, Drizzle ORM, Zod
API contract	OpenAPI-first; Orval codegen → React Query hooks + Zod schemas
Real-time	Shared HTTP + WebSocket server; REST mutations broadcast to clients
i18n	i18next
Auth	Clerk (<code>@clerk/expo</code> mobile, <code>@clerk/express</code> server)
Payments	Stripe + <code>stripe-replit-sync</code> (webhooks mirrored to Postgres)
Voice	ElevenLabs TTS (Expo Speech fallback)
Environment	OpenWeather API (server-proxied, cached)

Folder model (mobile): `app/` (routes/layouts), `components/`, `services/` (business logic), `store/` (reducer slices), `utils/` (pure helpers), `featureFlags/`, `theme/` (brand system), `design/` (Figma tokens), `data/` (SKUs, plans, templates), `types/`.

4. App Surfaces — Screens & Navigation

Navigation is **locked to 5 visible tabs**; additional surfaces exist as stacked or hidden routes (engines built, surfaces revealed over time).

Visible tabs: Home · Hydration (Journal) · Protocols · Community · Profile.

All routes (`app/`):

- Tabs: `index` (Home), `protocol`, `scan` (HydroScan), `journal`, `competition`, `social`, `sleep`, `profile`, `social-legacy` (*several hidden via `href:null`*).
- Auth: `(auth)/sign-in`, `(auth)/sign-up` (Clerk).
- Hidden / staged: `(hidden)/cruise/*` (`journey · port · excursion · recovery`), `circles/[id] · manage · shared`, `heat/guardian`, `ring/session`, `subscription/manage`, `legal/privacy · terms · health-disclaimer`, `clutch` (team command grid), `phantom` (wearable status), `territory` (performance map), `urine-check`, `onboarding`, `weekly-report`.

Home composition (current, post-revert orb-focused layout): readiness eyebrow → status headline → Status Pulse Orb → status label → live status line → consequence line → status-based primary CTA → quick-action grid → Hydration Status card → flag-gated zones (render nothing when off) → pinned “AFORCE” brand footer.

5. Feature Flags & Phased Exposure

Phase 1 (Core) is on by default; Phase 2/3 sub-features ship **off** and light up over time. Verified defaults from `featureFlags/flags.ts`:

On by default (`true`): `city_competition_enabled`, `state_competition_enabled`, `team_competition_enabled`, `global_leaderboard_enabled`, `cruise_mode_enabled`, `sleep_mode_enabled`, plus spec flags `spec_activation`,

`spec_social`, `spec_coachV2`, `spec_recoveryCircle`, `spec_notifications`, `spec_orb`, `spec_timelineLock`,
`spec_hydroJournal`, `spec_hydroScan`, `spec_profileSource`, `spec_sharedContextLayer`, `spec_uiFreeze`.

Off by default (false): `clutch_access_enabled` (+ heat/inventory/clip), `guardian_intelligence_enabled` (+ body_map/alerts),
`phantom_wearable_enabled`, `ring_enabled`, `kids_world_enabled`, `demo_mode_enabled`, `metabolic_readiness_enabled`
(+ glucose), `performance_age_enabled`, `voice_checkin_enabled`, `intent_capture_enabled`,
`performance_statements_enabled`, `offline_intake_outbox_enabled`, all `cruise_*` sub-features,
`voice_status_module_visible`, and gated locales `spec_language_ar/zh/ja/ko/hi`.

6. Hydration Scoring Engine

A single 0–100 score drives every coach surface in lock-step.

- **Event-driven scoring.** Each logged intake event has defined point values, **absorption caps** (per rolling time window), and **time-release curves** so a large gulp is absorbed gradually rather than all at once.
- **Continuous depletion.** A pure, dependency-free helper computes score-points-per-minute decay from physiological standards, so the score falls between logs.
- **Daily water target** is derived per user from body weight, activity, and climate (heat/humidity from the weather proxy), not a fixed number.
- **Score Protection** is enforced at the engine boundary: scans and recommendations are advisory and never mutate score; only completed intake/recovery behavior does. Scoring runs **on-device**; the server persists client-submitted snapshots rather than acting as the authority.
- **Modifiers:** Social/Night Mode (alcohol) raises decay; Heat Guard and low-score states apply small product-specific recovery bonuses.

(Exact point/cap constants live in `utils/scoringEngine.ts`, `services/sweatRateEngine.ts`, and the depletion helpers — parameterized in code, intentionally not frozen here.)

7. Performance-State Color System

Five score bands (single source of truth: `theme/statusColor.ts`, brand v2.1.0). Color is a **signal only** — borders, dots, glows, accents, CTA tint — never a fill.

Band	Range	Calm hex	Meaning
OPTIMAL	85–100	<code>#1FA35A</code> (Soursop green)	Peak — soft wide glow
STABLE	70–84	<code>#3DBE7A</code> (green, lighter)	Holding — subtle glow
DECLINING	50–69	<code>#FFDE00</code> (amber)	Slipping — minimal glow
RISK	30–49	<code>#FF8C1A</code> (orange)	Act now — medium glow
CRITICAL	0–29	<code>#FF2800</code> (signal red)	Depleted — tight intense glow

A **Pressure Mode** variant deepens saturation, raises glow alpha, and speeds the pulse for the same band.

This five-band ladder (`theme/statusColor.ts`) drives the **AI Coach status-color layer** — dots, borders, glows, CTA tint — and the score read-out. A **separate, intentional** four-band *Performance State* ladder (`utils/scoringEngine.ts` → `resolveState` : `PEAK ≥90` / `BALANCED ≥75` / `RECOVERING ≥60` / `DEPLETED`, with its own colors in `theme/colors.ts` `states`) drives the **orb** (pulse / flare-on-peak / collapse-on-depletion), `riskTimer`, and command selection. The two ladders use different mid-band thresholds by design but share the same top green (`#1FA35A`) and bottom red (`#FF2800`); see `design/aforce-design-tokens.md` for both tables.

8. Subscription Tiers

Verified monthly prices from `data/subscriptionPlans.ts`.

Consumer (5 tiers):

Plan	Price	Unlocks
Core	\$9.99/mo	Hydration Control Center, Status Pulse + score, basic protocol, basic AI, logging, smart reminders
Recovery+	\$9.99/mo	Adds post-session Recovery Mode
AForce Athlete	\$19.99/mo	Enhanced AI, personalized protocol, advanced recovery, 90-day trends, competition + city/state/team leaderboards, premium notifications, Metabolic Readiness
Performance Bundle	\$59.99/mo	Flagship: Athlete tier + monthly product drop (1 canister or 2 stick packs) at member pricing, priority AI/insights
AForce Elite	\$99/mo	Guardian Mode (individual), premium analytics, full monthly product bundle, early access, concierge support

Team (Core): Team Core Starter \$49/mo (≤25 seats) · Team Core Growth \$99/mo (≤50) · Team Core Pro \$149/mo (≤100) — roster-aware, group reporting, admin console.

Enterprise — Clutch (live team command): Clutch Starter \$1,000/mo · Clutch Pro \$2,500/mo · Clutch Elite \$5,000/mo.

Enterprise — Guardian (injury-risk protection): Guardian Core \$5,000/mo + \$7,500 setup (6-mo min) · Guardian Elite \$8,000/mo + \$12,500 setup (12-mo min).

Entitlement is gated client-side via `useEntitlement.ts`; Stripe is the source of truth, mirrored to Postgres.

9. Product Store (Physical SKUs)

Catalog model: one SKU per (format × flavor), plus per-format bundles (`data/pricing.ts` , `data/products.ts`). Subscriber/member pricing is lower than list. Pricing/shipping/tax are computed server-side.

- **Formats:** Stick packs (12 ct) · RTD cans (12 pk) · Canisters (30 servings).
- **Flavors:** Berry · Watermelon · Soursop.
- **Bundles:** flavor-agnostic multi-packs (flavor split chosen at checkout).
- Per-serving sodium and serving sizes are encoded per SKU for the sweat-rate math.

10. Data Model (PostgreSQL / Drizzle)

Table	Key columns
<code>aforce_user_state</code>	<code>user_id</code> (PK), <code>units_consumed_today</code> , <code>oz_target</code> , <code>symptom_state</code> , <code>biometrics</code> (JSONB), <code>social_mode</code> (JSONB), <code>intake_events</code> (JSONB)
<code>aforce_intake_logs</code>	<code>id</code> , <code>user_id</code> , <code>fluid_type</code> , <code>oz_amount</code> , <code>score_before</code> , <code>score_after</code>
<code>aforce_score_snapshots</code>	<code>id</code> , <code>user_id</code> , <code>score</code> , <code>level</code> , <code>recovery_score</code> , <code>captured_at</code>
<code>aforce_users</code>	<code>id</code> , <code>stripe_customer_id</code> , <code>plan_id</code> , <code>subscription_status</code> , <code>referral_code</code>
<code>aforce_circle_users</code>	<code>owner_user_id</code> , <code>member_user_id</code> , <code>status</code>
<code>aforce_analytics_events</code>	<code>event_id</code> , <code>analytics_id</code> , <code>event_type</code> , <code>payload</code>

11. API Surface (`api-server`)

Express 5 with Zod input/output validation generated from the OpenAPI spec.

Method · Path	Purpose
GET <code>/api/aforce/state</code>	Full <code>UserState</code> snapshot
POST <code>/api/aforce/intake</code>	Log fluid consumption
POST <code>/api/aforce/signals</code>	Update active symptoms (mild/moderate/severe)
GET <code>/api/aforce/journal/timeline</code>	Interleaved snapshots + intake history
POST <code>/api/aforce/social/activate</code>	Start a Social-Mode drinking session
POST <code>/api/aforce/social/cruise</code>	Start 24h Enterprise Cruise Mode
GET <code>/api/aforce/weather</code>	OpenWeather proxy (tempC, humidity) by lat/lon
POST <code>/api/voice/tts</code>	ElevenLabs text-to-speech proxy
POST <code>/api/checkout/session</code>	Stripe Checkout for subscriptions
POST <code>/api/stripe/webhook</code>	Stripe events → updates <code>plan_id</code>

Hardening: `SELECT ... FOR UPDATE` to serialize concurrent user actions; rate limiting on public + authenticated endpoints; in-memory TTL cache on the weather proxy; structured request/non-request logging.

12. Integrations

- **Auth — Clerk.** Custom email/password + Google SSO. JWT via `Authorization: Bearer` (REST) and `?token=` (WebSocket). Token bridging into the OpenAPI client; auth-gated routes on both mobile and server.
- **Real-time — WebSocket** mounted on the shared server (`/api/aforce/ws`) with periodic heartbeats; REST mutations are broadcast to connected clients.
- **Payments — Stripe** via `stripe-replit-sync`; the webhook secret is pulled from the managed Replit Stripe connector (not an env var). Webhook events mirror subscription state to Postgres.
- **Environment — OpenWeather**, server-proxied with a ~10-minute TTL cache (`OPENWEATHER_API_KEY`).
- **Voice — ElevenLabs** (`eleven_turbo_v2_5`); static phrases cached. Expo Speech is the fallback.

13. Localization

Launch set (6, in the selector): English · Spanish · French · German · Portuguese · Italian. Additional locales (Arabic, Chinese, Japanese, Korean, Hindi) are wired as resources but **gated behind flags** and excluded from the language selector. Architecture is modular so new languages add without a rebuild. (Number/date localization is a known gap — formatting is currently device-locale bound.)

14. Publishing & Distribution

- **Build service:** EAS Build — `development`, `preview`, `production` profiles in `eas.json`.
- **Bundle IDs:** iOS `com.aforce.os`, Android `com.aforce.os`.
- **Branding assets:** icon / splash / adaptive / favicon all render the **N–N “Non-Negotiable” monogram** (Bone `#F5F0E8` N’s, Signal Red `#C1281B` center bar, Cinematic Black `#0D0D0D`), generated from a font-free vector SVG.
- **Build:** `pnpm eas:build:ios` · `:android` · `:all` · `:preview` (from `artifacts/aforce-os/`).

- **Submit:** `pnpm eas:submit:ios` (needs ASC App ID + Apple Team ID via the `eas-configure-submit` helper) · `pnpm eas:submit:android` (needs a Google Play service-account JSON, never committed).
 - **Store screenshots** must be captured from a real device/simulator build — the web preview can't produce submission-grade assets.
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15. Design System (Brand v2.1.0)

- **Canvas:** near-black `#0D0D0D` . Content floats; soft radial glows, never hard box shadows. Generous spacing.
 - **Accents:** Signal Red `#C1281B` used sparingly (thin lines, eyebrows, active states, CTAs); Soursop green `#1FA35A` for positive status; Berry blue `#1E5BFF` for secondary data. Pure black `#000000` reserved for scrim/shadows/inverse text.
 - **Type (three roles):** Archivo Black (display) · IBM Plex Mono (eyebrows / metrics) · Inter (body).
 - **Tokens:** exported as `design/aforce-tokens.json` (Tokens Studio) + human-readable `design/aforce-design-tokens.md` . Served at `GET /api/design-tokens` and `GET /api/design-guide` .
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Source of truth: the AForce OS codebase. Locked product decisions are maintained in `replit.md` ; this spec mirrors them and adds the concrete, code-verified inventory.

AForce OS Constitution

Version: 1.0 **Date Locked:** July 3, 2026 **Owners:** Julius Burrell, Brandon Burrell **Status:** Locked

Engineering North Star

When engineering faces a trade-off between shipping quickly and preserving long-term trust, **long-term trust wins.**

Principles

1. **The body comes first. The recommendation comes second. The product comes last.**
 2. **One hero metric. HydroState.** Never create a competing hero metric.
 3. **Every recommendation must be explainable in plain language, using the user's own data.**
 4. **The OS learns the individual, not the population.** Personal baseline always outranks population average once sufficient individual data exists.
 5. **Observation, never diagnosis.** The OS notices patterns. It does not claim medical authority.
 6. **The OS speaks only when speaking adds value.** Silence is itself a form of trust.
 7. **Who sees this data, and why, must be answered before any feature ships.**
 8. **Build once. Reveal over time.** Architecture is built complete; surfaces release in phases behind feature flags.
 9. **Complete Architecture. Controlled Visibility.** AForce OS is built as one integrated operating system. Features may be hidden behind feature flags and phased releases, but the architecture is designed to work together from the beginning. New phases reveal capability — they do not require rebuilding the foundation.
 10. **AForce OS exists to earn trust, not attention.** Every recommendation must make the next recommendation more trusted than the last. If trust is ever sacrificed for engagement, engagement loses. Every time.
 11. **Never optimize for time spent in the app. Optimize for confidence outside the app.** Open app. Get command. Execute. Close app. Live life. Come back only when needed. (Stated exception: Guardian and Clutch, which serve a coach/staff safety relationship where active attention during risk windows is the explicit value delivered.)
 12. **AForce OS does not exist to tell people about their bodies.** It exists to learn this person so well that every future recommendation becomes more personal, more trusted, and more effortless than the last.
 13. **Your body teaches AForce OS.** AForce OS listens, remembers, and reflects what it learned back to you. It is never the one claiming to know best.
 14. **Every new feature must prove it improves the operating system after one year of use** — not just the first day of use.
 15. **Every conversation should make the next conversation shorter, more personal, and more useful.** The AI Coach is measured by reduction in required interaction over time, not by engagement volume. AForce OS is not a chatbot waiting for questions. It is a system that already understands the person well enough that fewer questions are ever necessary.
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The AForce Promise

GATE (must pass to be considered at all):

Does this help us understand this person better? If no — stop. Nothing else matters. Do not evaluate against the constraints below.

CONSTRAINTS (only apply if the gate is passed):

- Can it be explained in one sentence?
- Does it reduce effort instead of adding effort?
- Will this still matter five years from now?

A feature that fails the gate does not ship, regardless of how simple, effortless, or durable it appears to be.

The Three Question Rule

Applied before any feature ships, even if already designed:

1. **Does this make the OS know this specific person better?** If no, don't ship it.
2. **Does this make the user's day easier?** If no, don't ship it.
3. **Would the Founding 200 actually notice if we removed it?** If no, it probably shouldn't ship yet.

This third question exists specifically because founders often love features users barely notice. It keeps the team focused on delivering things that create obvious value.

Quarterly Review Practice

Before any new major feature is approved for the next quarter, ask:

“If we were starting AForce OS today, knowing everything we've learned, would we build this feature again?”

If no, remove it. This happens as part of the quarterly approval gate — not as a separate standing meeting that competes with shipping deadlines.

Learning Journal Protocol

All Founding 200 feedback goes into `Learning-Journal.md` first. Nothing from a single beta tester is ever written directly into this Constitution or the Architecture Appendix. Only after a pattern emerges across multiple users does it graduate into a proposed Architecture change — and only Julius and Brandon approve that graduation.

Change Control

No changes to this Constitution may be implemented without:

- A documented proposal
- Supporting beta evidence (never a single account or single conversation)
- Review by Julius and Brandon

- Updated version number

AForce OS — Architecture Appendix

Contains everything already built and specced. Every section carries a **Status** field.

Status options: Build Now · Architecture Only · Phase 2 · Phase 3 · Phase 4

Governing documents: [AForce-Constitution.md](#) and [AFORCE_OS_ARCHITECTURE_V1.md](#) . This appendix records build/phase status per section; it does not restate the full architecture.

HydroState™ (Sections 1–17)

Status: Build Now (core), Phase 2 (Visual Intelligence, gated by `hydrostate_visual_enabled` until device/lighting validation complete).

Sections 18–27 — Adaptive Profile & HydroState Intelligence

Adaptive Profile Engine, Performance Profile, Body Recalibration Engine, Sleep Readiness Intelligence, Tomorrow Load Forecast, Hydration Resilience Score, Oral Hydration Signal, Advanced Visual Intelligence, Personal Adaptive Learning, Performance Drift.

Status: Build Now (architecture), Phase 2 (surface).

HydroScan™ (Sections 28–37)

Performance Decision Intelligence, Scan Anything, Performance Fit, Decision Confidence, Strengths & Considerations, Recovery Intelligence, Decision Memory, Integration Rule, Trust Principle, Global HydroScan.

Status: Build Now (base scan). See **DR-001** for launch-scope amendments.

Launch amendment — DR-001 (2026-07-17), controlling for launch: §35 Integration Rule is **advisory-only** — HydroScan surfaces advisory signals and never mutates a score (Score-Protection isolation stays permanent). §32 / §33 / §34 / §29-OCR are **post-launch** (deferred, not defects); built-but-dark HydroScan 2.0 stays behind its flags, gated by CR-1. Full record: [governance/decisions/DR-001-hydroscan-integration-and-launch-scope.md](#) .

Sections 47–52 — Engagement, Sharing & Transparency

Performance Sharing, Performance Referral, Year in Performance, Performance Status, Privacy Center, Explainability Center.

Status: Phase 2.

Sections 53–57 — V1 Refinements

Data Freshness, Signal Quality, Profile Completeness, Universal Personalization, Performance & Battery Optimization.

Status: Build Now.

Section 58 — Command Confidence Display

Status: Build Now.

UI surfacing on the existing Signal Weighting Engine and Command Confidence. **No new calculation logic.** Applies to Today's Command, HydroScan Performance Fit, Recovery Window, Sun Recovery Mode.

Section 59 — Adaptive Response Engine (extends Decision Memory)

Status: Build Now.

Tracked categories: Heat, Hydration, Recovery, Sleep, Caffeine, Alcohol, Travel, Training, Cramp, Recovery Speed, Performance Consistency Response.

Includes Personal Response Library, What Worked, Confidence After Action.

Language rule: cause-and-effect only, never risk or diagnosis language. Never the words *risk*, *injury*, *diagnosis*, *prevent*. Recurring/severe symptoms always prompt physician consultation.

Section 60 — Response Timeline

Status: Phase 2.

Query layer on existing Performance Memory. Feature flag defaults **off** until 60–90 days of personal data exists for that user.

Section 61 — Living Performance Model

Status: Build Now (daily single-lesson surface only), Phase 2–4 (Your Body's Manual, Confidence Journey, Legacy summary).

User-facing language is always “**your body taught us**” — never “what did I learn about [name].” Long-term Legacy summaries never use prevention or causal medical language (“prevented X events”); always reflect completed behavior (“completed X commands, improved Y%, thank you for showing up”).

Silent Intelligence: when on track, say so plainly — “You're exactly where you should be.”

Section 62 — Founder Mode & Four-Environment Architecture

Status: Build Now, internal only, never exposed in Production.

Four environments:

- **Production** — real users, phased by feature flag, reads/writes Production only.
- **Founder Mode** — Julius, Brandon, internal team only. Full unrestricted access to every feature regardless of phase, including features mid-build with no phase number yet assigned. Reads Sandbox or Production. Writes Sandbox only.
- **Engineering Sandbox** — infrastructure layer, separate responsibility from Founder Mode interface. Writes Sandbox only.
- **Demo Mode** — investors, scripted, reads seeded demo data only (`data/demoProfile.ts`), writes nothing.

Founder Control Center (built as part of Founder Mode):

- Instant per-feature toggle on/off
- View as Phase 1 / Phase 2 / Phase 3 / Phase 4
- Simulate new user / 30 days / 90 days / 1 year of usage (Time Travel)
- Simulate different user types — beta user, coach, parent, enterprise customer
- Compare versions side by side before release decisions

- Scenario Engine — pre-built real usage scenarios, at least one sourced from a real documented competitor failure (e.g. recurring cramping where sodium-only replacement failed), not only idealized personas

Isolation: separate database schema for Sandbox, persistent visual watermark (**FOUNDER MODE / SANDBOX**) whenever active. No additional overlapping safeguards beyond this unless a specific demonstrated problem requires it.

Section 63 — Guardian, Clutch, Cruise Mode Compliance Pass

Status: Build Now (required revisions to existing specs, not new features).

- Profile recalibration always framed as performance optimization, never correction, never used for urgency/re-engagement.
- Guardian and Clutch are a stated exception to Principle 11 (trust over attention) — they serve a coach/staff safety relationship where active attention during risk windows is the explicit value delivered. This exception applies only to Guardian and Clutch.
- Cruise Mode streak mechanic requires revision: replace do-not-break-the-chain framing with the **Resolved Today / Carrying Forward** framing already proven in Evening Debrief. Never threaten loss of a streak if a day is missed.

Section 64 — Conversational Intelligence Architecture™

Status: Build Now.

Governs the AI Coach behavior layer (`services/voiceService.ts` and any future conversational surface).

AForce OS is not an AI chatbot. The AI does not exist to answer questions. It exists to understand the person over time so well that fewer questions are ever necessary.

Core behavior rules:

1. The AI does not wait passively for input. It already knows enough about the user's current HydroState, Performance Memory, and recent patterns to speak first when speaking adds value — and to stay silent when it does not (per Constitution Principle 6).
2. Every conversation should make the next conversation shorter, not longer. A returning user should never have to re-explain context the OS should already know.
3. The AI earns trust by remembering, adapting, and speaking only when it adds value — never by being talkative, never by maximizing engagement through conversation volume.
4. Success is measured by reduction, not expansion: fewer clarifying questions needed over time, shorter time-to-command, less back-and-forth required for the user to get what they need.
5. When a user initiates a conversation, the AI always responds with full context already loaded — current HydroState, recent Adaptive Response patterns, active Recovery Window status, relevant Personal Response Library entries — never starting from zero.
6. The AI Coach speaks in the locked voice per coach persona (Rock, BB, Surge, Sage) and in the observation-only language already locked across every other module — markers suggest, never diagnosis, cause-and-effect from Personal Response Library, never population comparison.

What this means in practice: a standard AI assistant waits for a question, asks clarifying questions, then answers. AForce OS already initiates the relevant command before the user needs to ask, because HydroState, the Adaptive Response Engine, and Environmental Pressure already combined to determine the answer. If the user does ask anyway, the AI already has every relevant signal loaded and answers immediately, in one exchange, using the user's own history.

This section does not require new data architecture. It requires the existing voice and chat layer to be built against these behavior rules from the start, rather than defaulting to a standard question-and-answer chatbot pattern.

AForce OS — Phase Roadmap

Phase status per section is tracked in [Architecture-Appendix.md](#). This roadmap groups the phased rollout.

Phase 0 — Founder Build

Full architecture built. Nothing exposed publicly. Julius and Brandon have complete access via Founder Mode to every feature regardless of phase.

Phase 1 — Founding 200 (Free)

HydroState core, Today's Command, Evidence Engine, Performance Memory, basic Voice Coach (built per Section 64 Conversational Intelligence rules), basic HydroScan, Adaptive Response Engine, Command Confidence Display, Living Performance Model (daily lesson only), Climate Profile, Environmental Pressure, Sections 53–57 optimizations, Section 63 compliance revisions.

Phase 2 — Subscription tiers activate

HydroState Visual Intelligence (after device/lighting validation), Sleep Readiness Intelligence, Tomorrow Load Forecast, Response Timeline, Your Body's Manual, Confidence Journey, full Social layer, Performance Sharing, Performance Referral, full HydroScan intelligence, Sun Exposure System.

Phase 3 — Phantom Band + Enterprise

Phantom Band Core hardware, Performance Age full activation, Command Confidence full intelligence, Clutch, Guardian, Cruise Mode (with revised streak mechanic), Meridian luxury tier.

Phase 4 — Long-horizon features

Legacy summary, full global localization, advanced Adaptive Response learning, future sensors and intelligence modules.

Final Rules for Claude Code

The operating contract for building the AForce OS governance sections. Read alongside [AForce-Constitution.md](#), [Architecture-Appendix.md](#), and [Phase-Roadmap.md](#).

1. Create the `/governance` folder structure first, before building any of Sections 58–64.
2. Build one numbered section at a time. Confirm completion before starting the next.
3. Test each section before moving to the next.
4. Every section in `Architecture-Appendix.md` must carry its Status tag — Build Now, Architecture Only, Phase 2, Phase 3, or Phase 4.
5. Section 58 is UI only — no new calculation engine.
6. Section 59 extends existing Decision Memory — do not build a parallel system. Language rule is non-negotiable: cause-and-effect only, never risk/diagnosis language.
7. Section 60 stays feature-flagged off until 60–90 days of personal data exists per user.
8. Section 61 daily lesson ships now; Your Body's Manual, Confidence Journey, and Legacy summary are architecture-only until their assigned phase. "Your body taught us" is a hard compliance rule, not a style preference. Legacy summaries never use prevention/causal medical language.
9. Section 62 Founder Mode is internal-only, never exposed in Production, accessible only to authenticated Julius/Brandon/internal team accounts. Writes only to Sandbox.
10. Section 63 compliance revisions apply to the existing Profile, Guardian, Clutch, and Cruise Mode specs — required updates, not new features.
11. Section 64 governs all AI Coach and conversational behavior from Phase 1 onward. The AI Coach must be built against these rules from the start — it does not default to a standard chatbot pattern and get corrected later.
12. The AForce Constitution is frozen after this email. No further philosophical principles without explicit approval from both Julius and Brandon, and only after real-world beta evidence.
13. All thresholds, weights, and tunable values continue to live in `config/hydroStateModel.ts`. Never hardcoded.
14. No redesign of existing navigation. No new tabs. No rebuild of existing systems.